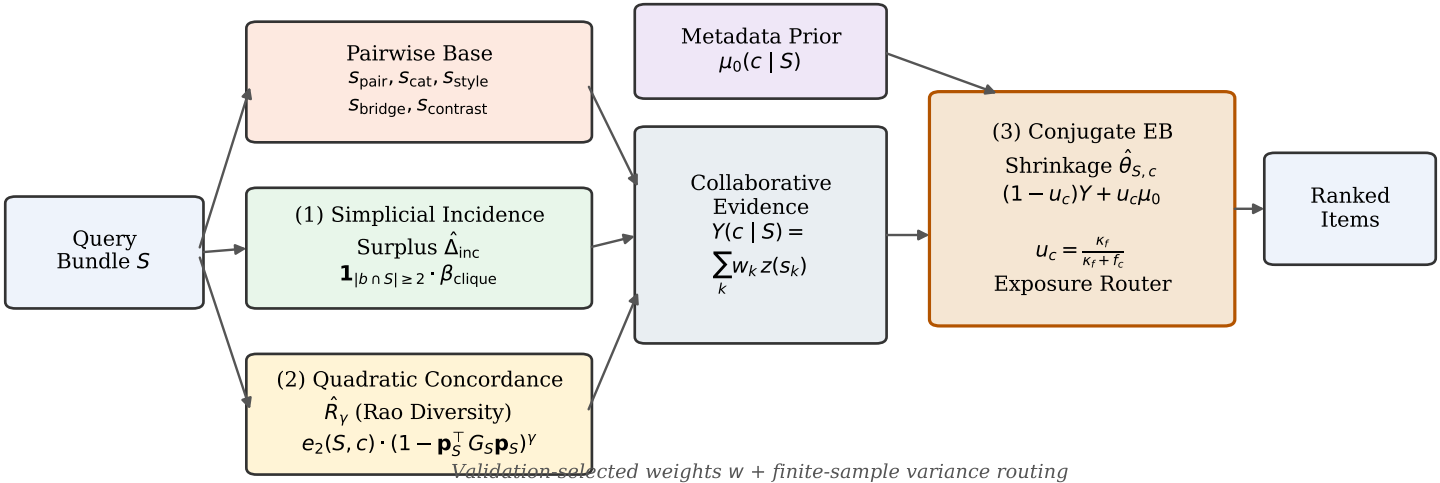
Query Bundle S
Pairwise Base
$s_{\text{pair}}, s_{\text{cat}}, s_{\text{style}}$
$s_{\text{bridge}}, s_{\text{contrast}}$
(1) Simplicial Incidence
Surplus $\hat{\Delta}_{\text{inc}}$
$\mathbf{1}_{|b \cap S| \geq 2} \cdot \beta_{\text{clique}}$
(2) Quadratic Concordance
$\hat{R}_\gamma$ (Rao Diversity)
$e_2(S,c) \cdot (1 - \mathbf{p}_S^\top G_S \mathbf{p}_S)^\gamma$
Metadata Prior
$\mu_0(c \mid S)$
Collaborative Evidence
$Y(c \mid S) = \sum_k w_k z(s_k)$
(3) Conjugate EB
Shrinkage $\hat{\theta}_{S,c}$
$(1-u_c)Y + u_c\mu_0$
$u_c = \frac{\kappa_f}{\kappa_f + \hat{f}_c}$
Exposure Router
Ranked Items
Validation-selected weights w + finite-sample variance routing